Bansal Group of Institutions

Department of Training

Certificate Courses (for all branches) - I Sem : Software Domain + Communication Skill

CERTIFICATION COURSES_TCS ION, HP, HCL,CISCO, SAYLOR, COURSERA ETC. (I SEM)

Note: The courses/assessment available in different platforms(free/paid) are listed here as suggestion.Other courses(important from placement/skill point of view) can also be incorporated.

NAME OF COURSES	Duration	Cost	About the Course	Semester	Category	Platform	Link
COMM001: Principles of Human Communication	56 hrs	Free	Learn fundamental communication principles, common communication practices, and communication theories that will help you better understand the communication transactions that you experience in your daily life. This course will introduce you to communication principles, common communication practices, and a selection of theories to better understand the communication transactions that you experience in your daily life.	I Sem	Comm Skill	**Saylor**	https://learn.saylor.org/?utm_source=hp&utm_medium=CTA&utm_campaign=Start
COMM101: Public Speaking	92 hrs	Free	Explore the elements of an effective speech and practice specific, performance-oriented aspects of public speaking This course systematically examines the elements of an effective speech and goes through an element-by-element examination of the essentials of public speaking while also identifying traits of the individual speaker and how they affect preparation and presentation. This course also demonstrates specific, performance-oriented aspects of public speaking.	I Sem	Comm Skill	**Saylor**	https://learn.saylor.org/?utm_source=hp&utm_medium=CTA&utm_campaign=Start
PRDV003: Word Processing	4 hrs	Free	Get an introduction to word processing using Microsoft Word and Google Docs in this course for people with little or no word processing experience. Explore word processing skills by creating common business documents. This course will teach you how to create documents using Microsoft Word and Google Docs. It is designed for students with little or no experience with these word processing programs. We will review some basic editing and formatting tools and how to create and print a business letter. Google Docs and Microsoft Word share many of the same formatting features and functions, and most of the features we explore in this course can be used in both programs. If you do not have Microsoft Word, you can use a free 30-day trial of Microsoft Office 365. Google Docs is available for free to anyone with a Google account. Many schools, universities, and public libraries also have at least one word processing program that you can use.	I Sem	Professional Development	**Saylor**	https://learn.saylor.org/?utm_source=hp&utm_medium=CTA&utm_campaign=Start
PRDV004: Spreadsheets	4 hrs	Free	Get a comprehensive introduction to spreadsheets in this course that covers everything first-time users need to know. This course is a comprehensive introduction to Microsoft Excel and Google Sheets. We will discuss spreadsheets, their terminology, and their components, and explore how to use a spreadsheet in your everyday life at work, school, and home.	I Sem	Professional Development	**Saylor**	https://learn.saylor.org/?utm_source=hp&utm_medium=CTA&utm_campaign=Start
PRDV103: Interviewing Skills	04 hours	Free	Learn how to showcase your personality, strengths, interests, and abilities to potential employers through the interview process. This course will help you showcase your knowledge, skills, abilities, strengths, and interests to potential employers during a professional job interview. Be sure to review our other professional development courses for more on other skills that will help you get ahead, such as preparing a resume, interpersonal communication, professional etiquette, leadership, teamwork, and understanding organizational culture.	I Sem	Professional Development	**Saylor**	https://learn.saylor.org/?utm_source=hp&utm_medium=CTA&utm_campaign=Start
PRDV213: Group Communication	07 hours	Free	Learn the principles of small group communication that will help you function more effectively in groups, both in your personal life and in the workplace. This course will focus on the principles of small group communication to help us better navigate this important communication setting.	I Sem	Professional Development	**Saylor**	https://learn.saylor.org/?utm_source=hp&utm_medium=CTA&utm_campaign=Start
Introduction to Soft Skills	Duration: 1 Weeks Commitment: 2 Hours	Free		I Sem	Comm Skill	**TCS**	https://www.tcsion.com/hub/career-creator/free-learning-courses/
Communication Skill	No more than 60 minutes	Free	In this course, you will learn how to: Determine what information in a presentation is most relevant to a specific audience. Create effective and well-designed slide presentations to be given remotely or in person. Develop best practices for public speaking and giving a presentation.	I Sem	Comm Skill	**HP**	https://www.life-global.org/allcourses?page=1
Get Connected	30 hours	Free*	Core Skills: Create accounts for Facebook, LinkedIn and YouTube. Understand computer networking and how to browse and search the Internet. Become familiar with Microsoft Windows and how to work with files and folders. Identify common problems and implement solutions.	I Sem	Software	**CISCO**	https://www.cisco.com/site/us/en/learn/training-certifications/index.html
CLA: Programming Essentials in C	70 hours	Determined by Institution	Core Skills: Understand common computer programming concepts and write your own programs. Learn the syntax, semantics and basic data types of C. Apply programming skills using hands-on lab activities. Prepare for the C Programming Language Certified Associate Certification (CLA).	I Sem	Software	**CISCO**	https://www.cisco.com/site/us/en/learn/training-certifications/index.html
CPA: Programming Essentials in C++	70 hours	Free*	Core Skills: Learn the syntax, semantics, and basic data type of C++. Understand principles of the object-oriented model. Implement and write in C++ and resolve typical implementation challenges via language libraries. Prepare for the C++ Certified Associate Programmer Certification (CPA).	I Sem	Software	**CISCO**	https://www.cisco.com/site/us/en/learn/training-certifications/index.html
Introduction to Microsoft Excel	1.5 hours	Free	Learn step-by-step In a video that plays in a split-screen with your work area, your instructor will walk you through these steps: • The Microsoft Office365 Excel Program and view the basics of Excel (6 min) • Learn the layout, navigation, and ribbon (7 min) • Create a spreadsheet of information you are tracking (5 min) • Create basic formulas to analyze your data (6 min) • Format your data (6 min)	I Sem	Software	**Google**	https://grow.google/intl/en_in/certificates/
C# (Basic)	1 hr assessment	Free	Covers topics like Exception Handling, C# Program Structure, Properties and Indexers, Types and Variables	I Sem	Software	**Hacker Rank**	https://www.hackerrank.com/skills-verification

Bansal Group of Institutions
Department of Training

Certificate Courses (for all branches) - II Sem : Software Domain + Communication Skill

CERTIFICATION COURSES_TCS ION, HP, HCL,CISCO, SAYLOR, COURSERA ETC. (II SEM)

Note: The courses/assessment available in different platforms(free/paid) are listed here as suggestion.Other courses(important from placement/skill point of view) can also be incorporated

NAME OF COURSES	Duration	Cost	About the Course	Semester	Category	Platform	Link
ESL001: Elementary English as a Second Language	18 hours	Free	Improve your English! Learn new words and how to use active reading, grammar, and writing strategies to read and write English better at the A2 level. Unit 1, you will learn about the eight parts of speech and when each is used. You will learn how to recognize each part of speech and why using them correctly improves your communication skills. Units 2 and 3, you will learn how to read actively by using context clues and other reading skills and see how reading more can help you expand your vocabulary. Unit 4 focuses on taking those reading skills and using them to help you write. By the end of ESL001, you'll be able to write a few sentences about your daily routine using the lessons on parts of speech, reading actively, and vocabulary to help you communicate clearly. Successful completion of ESL001 will prepare you for ESL002, where you will continue to improve your English reading and writing skills at the A2 level.	I Sem	Comm Skill	**Saylor**	https://learn.saylor.org/?utm_source=hp&utm_medium=CTA&utm_campaign=Start
PRDV102: Resume Writing	5 hrs	Free	Develop the application materials you need for today's job market by honing your résumé writing skills, learning how to create an impressive résumé or improve the one you already have, and get suggestions for writing an effective cover letter. In this course, we explore techniques to help you construct a résumé and cover letter to help you stand out among other job candidates. We study several formats you can apply to various situations and help you assess your professional skills so you can be more competitive in today's job market. Finally, we discuss how to turn an average résumé and cover letter into a polished and effective marketing tool.	I Sem	Professional Development	**Saylor**	https://learn.saylor.org/?utm_source=hp&utm_medium=CTA&utm_campaign=Start
TCS iON Career Edge - Young Professional	Duration: 2 Weeks Commitment: 7-10 Hours/Week	Free	Covering key behavioural & communication skills and foundational skills in Accounting, IT and Artificial Intelligence, the learning in TCS iON Career Edge - Young Professional is delivered over 14 modules. Each module is of 1-2 hours duration with aids comprising videos, presentations, reading material, recorded webinars by TCS experts and end-of-module self-assessments that make it relevant and aligned to industry. The course has an end-of-course assessment that evaluates learners' understanding of concepts and if the course objectives are met, with the learner receiving a personalised certificate on successful completion. In addition, learners have access to a moderated digital discussion room that allows them to post queries and share their suggestions. The moderator responds to questions posted within a short time. Moreover, the course can be accessed anytime, anywhere, and on any device enabling Learning on the Go! TCS iON Career Edge - Young Professional enhances the employability quotient of youth in the new normal enabling then to make the transition from Campus to Corporate effectively.	I Sem	Comm Skill	**TCS**	https://www.tcsion.com/hub/career-creator/free-learning-courses/
Impress with Your Interview Skills	Duration: 1 Weeks	Free		I Sem	Comm Skill	**TCS**	https://www.tcsion.com/hub/career-creator/free-learning-courses/
Communication Skill	No more than 60 minutes	Free	In this course, you will learn how to: Determine what information in a presentation is most relevant to a specific audience. Create effective and well-designed slide presentations to be given remotely or in person. Develop best practices for public speaking and giving a presentation.	I Sem	Comm Skill	**HP**	https://www.life-global.org/allcourses?page=1
CPA: Programming Essentials in C++	70 hours	Free*	Core Skills: Learn the syntax, semantics, and basic data type of C++. Understand principles of the object-oriented model. Implement and write in C++ and resolve typical implementation challenges via language libraries. Prepare for the C++ Certified Associate Programmer Certification (CPA).	I Sem	Software	**CISCO**	https://www.cisco.com/site/us/en/learn/training-certifications/index.html
Introduction to Microsoft Excel	1.5 hours	Free	Learn step-by-step In a video that plays in a split-screen with your work area, your instructor will walk you through these steps: • The Microsoft Office365 Excel Program and view the basics of Excel (6 min) • Learn the layout, navigation, and ribbon (7 min) • Create a spreadsheet of information you are tracking (5 min) • Create basic formulas to analyze your data (6 min) • Format your data (6 min)	I Sem	Software	**Google**	https://grow.google/intl/en_in/certificates/

Bansal Group of Institutions

Department of Training

Certificate Courses (for CSE / IT / EC / EX) - III Sem : Software Domain + Communication Skill

CERTIFICATION COURSES_TCS ION, HP, HCL,CISCO, SAYLOR, COURSERA ETC. (III SEM)

Note: The courses/assessment available in different platforms(free/paid) are listed here as suggestion.Other courses(important from placement/skill point of view) can also be incorporated

NAME OF COURSES	Duration	Cost	About the Course	Semester	Category	Platform	Link
COMM001: Principles of Human Communication	56 hrs	Free	Learn fundamental communication principles, common communication practices, and communication theories that will help you better understand the communication transactions that you experience in your daily life. This course will introduce you to communication principles, common communication practices, and a selection of theories to better understand the communication transactions that you experience in your daily life.	III Sem	Comm Skill	**Saylor**	https://learn.saylor.org/?utm_source=hp&utm_medium=CTA&utm_campaign=Start
COMM101: Public Speaking	92 hrs	Free	Explore the elements of an effective speech and practice specific, performance-oriented aspects of public speaking This course systematically examines the elements of an effective speech and goes through an element-by-element examination of the essentials of public speaking while also identifying traits of the individual speaker and how they affect preparation and presentation. This course also demonstrates specific, performance-oriented aspects of public speaking.	III Sem	Comm Skill	**Saylor**	https://learn.saylor.org/?utm_source=hp&utm_medium=CTA&utm_campaign=Start
PRDV003: Word Processing	4 hrs	Free	Get an introduction to word processing using Microsoft Word and Google Docs in this course for people with little or no word processing experience. Explore word processing skills by creating common business documents. This course will teach you how to create documents using Microsoft Word and Google Docs. It is designed for students with little or no experience with these word processing programs. We will review some basic editing and formatting tools and how to create and print a business letter. Google Docs and Microsoft Word share many of the same formatting features and functions, and most of the features we explore in this course can be used in both programs. If you do not have Microsoft Word, you can use a free 30-day trial of Microsoft Office 365. Google Docs is available for free to anyone with a Google account. Many schools, universities, and public libraries also have at least one word processing program that you can use.	III Sem	Professional Development	**Saylor**	https://learn.saylor.org/?utm_source=hp&utm_medium=CTA&utm_campaign=Start
PRDV004: Spreadsheets	4 hrs	Free	Get a comprehensive introduction to spreadsheets in this course that covers everything first-time users need to know. This course is a comprehensive introduction to Microsoft Excel and Google Sheets. We will discuss spreadsheets, their terminology, and their components, and explore how to use a spreadsheet in your everyday life at work, school, and home.	III Sem	Professional Development	**Saylor**	https://learn.saylor.org/?utm_source=hp&utm_medium=CTA&utm_campaign=Start
PRDV102: Resume Writing	5 hrs	Free	Develop the application materials you need for today's job market by honing your résumé writing skills, learning how to create an impressive résumé or improve the one you already have, and get suggestions for writing an effective cover letter. In this course, we explore techniques to help you construct a résumé and cover letter to help you stand out among other job candidates. We study several formats you can apply to various situations and help you assess your professional skills so you can be more competitive in today's job market. Finally, we discuss how to turn an average résumé and cover letter into a polished and effective marketing tool.	III Sem	Professional Development	**Saylor**	https://learn.saylor.org/?utm_source=hp&utm_medium=CTA&utm_campaign=Start
PRDV103: Interviewing Skills	04 hours	Free	Learn how to showcase your personality, strengths, interests, and abilities to potential employers through the interview process. This course will help you showcase your knowledge, skills, abilities, strengths, and interests to potential employers during a professional job interview. Be sure to review our other professional development courses for more on other skills that will help you get ahead, such as preparing a resume, interpersonal communication, professional etiquette, leadership, teamwork, and understanding organizational culture.	III Sem	Professional Development	**Saylor**	https://learn.saylor.org/?utm_source=hp&utm_medium=CTA&utm_campaign=Start
PRDV213: Group Communication	07 hours	Free	Learn the principles of small group communication that will help you function more effectively in groups, both in your personal life and in the workplace. This course will focus on the principles of small group communication to help us better navigate this important communication setting.	III Sem	Professional Development	**Saylor**	https://learn.saylor.org/?utm_source=hp&utm_medium=CTA&utm_campaign=Start
TCS iON Career Edge - Young Professional	Duration: 2 Weeks Commitment: 7-10 Hours/Week	Free	Covering key behavioural & communication skills and foundational skills in Accounting, IT and Artificial Intelligence, the learning in TCS iON Career Edge - Young Professional is delivered over 14 modules. Each module is of 1-2 hours duration with aids comprising videos, presentations, reading material, recorded webinars by TCS experts and end-of-module self-assessments that make it relevant and aligned to industry. The course has an end-of-course assessment that evaluates learners' understanding of concepts and if the course objectives are met, with the learner receiving a personalised certificate on successful completion. In addition, learners have access to a moderated digital discussion room that allows them to post queries and share their suggestions. The moderator responds to questions posted within a short time. Moreover, the course can be accessed anytime, anywhere, and on any device enabling Learning on the Go! TCS iON Career Edge - Young Professional enhances the employability quotient of youth in the new normal enabling then to make the transition from Campus to Corporate effectively.	III Sem	Comm Skill	**TCS**	https://www.tcsion.com/hub/career-creator/free-learning-courses/
Impress with Your Interview Skills	Duration: 1 Weeks	Free		III Sem	Comm Skill	**TCS**	https://www.tcsion.com/hub/career-creator/free-learning-courses/
Communication Skills	Duration: 1 Weeks Commitment: 2 Hours	Free	About this Course Communication is as old as the existence of human beings. Words are not the only way of getting your message across. Many times you can sense the mood your colleague or friend is in. How do you think you can figure it out? It is because communication involves not just words, but also actions, body language and facial expressions to mention a few. In this course we will help you understand what communication is and why we communicate. You will also understand non-verbal communication in detail. On completion of the course, you will learn how to communicate effectively using a combination of verbal and non-verbal communication skills. By purchasing or subscribing to this program, you agree to abide by the Usage Policy, Privacy Policy, Security Policies and any Terms and Conditions mentioned on this portal.	III Sem	Comm Skill	**TCS**	https://www.tcsion.com/hub/career-creator/free-learning-courses/
Introduction to Soft Skills	Duration: 1 Weeks Commitment: 2 Hours	Free		III Sem	Comm Skill	**TCS**	https://www.tcsion.com/hub/career-creator/free-learning-courses/
Communication Skill	No more than 60 minutes	Free	In this course, you will learn how to: Determine what information in a presentation is most relevant to a specific audience. Create effective and well-designed slide presentations to be given remotely or in person. Develop best practices for public speaking and giving a presentation.	III Sem	Comm Skill	**HP**	https://www.life-global.org/allcourses?page=1
CS101: Introduction to Computer Science I	52 hours	Free	Explore the basic concepts, nomenclature, and historical perspective of computers and computing, and principles of software development and Object-Oriented Programming. This course will introduce you to the field of computer science and the fundamentals of computer programming. CS101 is specifically designed for students with no prior programming experience and touches upon a variety of fundamental topics. This course uses Java to demonstrate those topics. Java is a high-level, portable, and well-constructed computer programming language developed by Sun Microsystems (now Oracle).	III Sem	Software	**Saylor**	https://learn.saylor.org/?utm_source=hp&utm_medium=CTA&utm_campaign=Start

CS102: Introduction to Computer Science II	51 hours	Free	Explore this detailed survey of computing and programming, with an emphasis on understanding object-orientation and the Java and C++ computer programming languages. We will use history, theory, and practice to deliver lessons that prepare you for a career in computer science. We will begin with a comparison between Java, the programming language used in the previous course, and C++, another popular, industry-standard programming language. We will then discuss the fundamental building blocks of Object-Oriented Programming, reviewing what we have already learned while familiarizing ourselves with more advanced programming concepts. The remaining course units will be devoted to various topics, including the Standard Template Library, containers, exceptions, recursion, searching and sorting, and generic programming. By the end of the class, you will have a solid understanding of Java and C++ programming, as well as a familiarity with the major issues that programmers routinely address in a professional setting.	III Sem	Software	**Saylor**	https://learn.saylor.org/?utm_source=hp&utm_medium=CTA&utm_campaign=Start
CS401: Operating Systems	120 hours	Free	Examine how operating systems and design have evolved as changes in hardware and software led to contemporary operating systems. Topics include basic OS concepts, methods of OS design and construction, process coordination, management, and algorithms for CPU scheduling, memory, and general resource allocation. This course will introduce you to modern operating systems. We will focus on UNIX-based operating systems, though we will also learn about alternative operating systems, including Windows. The course will begin with an overview of the structure of modern operating systems. Over the course of the subsequent units, we will discuss the history of modern computers, analyze in detail each of the major components of an operating system (from processes to threads), and explore more advanced topics in the field, including memory management and file input/output. The class will conclude with a discussion of various system-related security issues.	III Sem	Software	**Saylor**	https://learn.saylor.org/?utm_source=hp&utm_medium=CTA&utm_campaign=Start
PRDV410: Introduction to Java and Object-Oriented Programming	6 hours	Free	Learn the basics of object-oriented (OO) programming, a paradigm that uses objects that contain data in the form of instance variables and methods that control the behavior of the objects. This course is suitable for anyone and uses Java to demonstrate key OO concepts. We will begin this course with an introduction to the difference between traditional structural program design and object-oriented (OO) programming. This course will explore the unique, dynamic approach that OO offers and discuss why it is an effective and flexible paradigm. We then move into the essential concepts that are part of all OO programming, including inheritance, polymorphism, abstract classes, encapsulation, and coupling.	III Sem	Software	**Saylor**	https://learn.saylor.org/?utm_source=hp&utm_medium=CTA&utm_campaign=Start
PRDV401: Introduction to JavaScript I	17 hours	Free	Learn fundamental programming concepts using JavaScript to produce code for browser-based interactive experiences using conditions and events. This course is the first in a series of three short courses that will build your programming skills in the JavaScript language. JavaScript is a client-side scripting language for producing interactive content in a web browser. It is one of the most popular programming languages worldwide. This course is for future JavaScript developers with some HTML and CSS knowledge. This course with a brief history of JavaScript. We then introduce programming concepts such as variables, control statements, objects, exception handling, and best practices. You will also learn about the Document Object Model or DOM, which provides the structure for dynamically displaying and manipulating elements on a webpage written in HTML using JavaScript.	III Sem	Software	**Saylor**	https://learn.saylor.org/?utm_source=hp&utm_medium=CTA&utm_campaign=Start
Introduction to IoT	20 hours	Free*	Core Skills: Learn how the current digital transformation is creating unprecedented economic opportunity. Understand how the IoT is bridging the gap between operational and information technology systems. Discover how standard business processes are being transformed. The security concerns that must be considered when implementing IoT solutions.	III Sem	Software	**CISCO**	https://www.cisco.com/site/us/en/learn/training-certifications/index.html
IoT Fundamentals: Connecting Things	70 hours	Determined by Institution	Core Skills: Learn how the current digital transformation is creating unprecedented economic opportunity. Understand how the IoT is bridging the gap between operational and information technology systems. Develop critical thinking and problem-solving skills using both real equipment and Cisco Packet Tracer. Soft skills such as teamwork and articulating problems and solutions in a business context.	III Sem	Software	**CISCO**	https://www.cisco.com/site/us/en/learn/training-certifications/index.html
Get Connected	30 hours	Free*	Core Skills: Create accounts for Facebook, LinkedIn and YouTube. Understand computer networking and how to browse and search the Internet. Become familiar with Microsoft Windows and how to work with files and folders. Identify common problems and implement solutions.	III Sem	Software	**CISCO**	https://www.cisco.com/site/us/en/learn/training-certifications/index.html
CLA: Programming Essentials in C	70 hours	Determined by Institution	Core Skills: Understand common computer programming concepts and write your own programs. Learn the syntax, semantics and basic data types of C. Apply programming skills using hands-on lab activities. Prepare for the C Programming Language Certified Associate Certification (CLA).	III Sem	Software	**CISCO**	https://www.cisco.com/site/us/en/learn/training-certifications/index.html
CPA: Programming Essentials in C++	70 hours	Free*	Core Skills: Learn the syntax, semantics, and basic data type of C++. Understand principles of the object-oriented model. Implement and write in C++ and resolve typical implementation challenges via language libraries. Prepare for the C++ Certified Associate Programmer Certification (CPA).	III Sem	Software	**CISCO**	https://www.cisco.com/site/us/en/learn/training-certifications/index.html
Introduction to Microsoft Excel	1.5 hours	Free	Learn step-by-step In a video that plays in a split-screen with your work area, your instructor will walk you through these steps: • The Microsoft Office365 Excel Program and view the basics of Excel (6 min) • Learn the layout, navigation, and ribbon (7 min) • Create a spreadsheet of information you are tracking (5 min) • Create basic formulas to analyze your data (6 min) • Format your data (6 min)	III Sem	Software	**Google**	https://grow.google/intl/en_in/certificates/
HTML, CSS, and Javascript for Web Developers	8 hours (approximately)	free	This course is part of the HTML, CSS, and Javascript for Web Developers Specialization When you enroll in this course, you'll also be enrolled in this Specialization. Learn new concepts from industry experts Gain a foundational understanding of a subject or tool Develop job-relevant skills with hands-on projects Earn a shareable career certificate	III Sem	Software	**Google**	https://grow.google/intl/en_in/certificates/
Introduction to Artificial Intelligence (AI)	8 hours 8 minutes	Free	What is AI? Applications and Examples of AI This week, you will learn what AI is. You will understand its applications and use cases and how it is transforming our lives. AI Concepts, Terminology, and Application Areas This week, you will learn about basic AI concepts. You will understand how AI learns, and what some of its applications are. AI: Issues, Concerns and Ethical Considerations AI is everywhere, transforming the way we live, work, and interact – that's why it's so important to build and use it in line with ethical expectations. This week, you will learn about issues and concerns surrounding AI, as well as how AI ethics helps practitioners build and use AI responsibly. This information will help you understand AI's potential impacts on society so that you can have an informed discussion about its risks and benefits. The Future with AI, and AI in Action This week, you will learn about the current thinking on the future with AI, as well as hear from experts about their advice to learn and start a career in AI. You will also demonstrate AI in action by utilizing Computer Vision to classify images.	III Sem	Software	**IBM**	https://www.classcentral.com/course/introduction-to-web-development-with-html-css-jav-53747
MySQL 8.0 Database Developer Exam Number: 1Z0-909	90 Minutes	Free	Review exam topics Connectors and APIs Choose between connectors for a given application, Demonstrate connector use, management and configuration, Retrieve data from the database by using a connector, Handle special values, Secure credentials and connections, Data-driven Applications Use prepared statements, Set SQL Modes to change MySQL behavior, Handle and interpret errors and warnings, Display query results, Aggregate and summarize data, Advanced Report Generation, MySQL Schema Objects and Data Design, create, and alter views, Store and process string data, Store and process numeric data, Store and process temporal data, Store and process spatial data, Transactions Control transactions in SQL, Control transactions in applications, Resolve consistency problems with isolation levels, Understand locking mechanisms within MySQL, Query Optimization Use indexes for optimization, Analyze queries for optimization, Rewrite queries for optimization, MySQL Stored Programs, Create and execute stored routines, Explain the programming constructs used in stored routines, Create and execute triggers, Schedule database operations, JSON and Document Store Create and store JSON documents, Process data in JSON documents, Explain application development with NoSQL and XDevAPI, Create and access a document store, Use MySQL Shell to access document stores	III Sem	Software	**Oracle**	https://mylearn.oracle.com/ou/home
Oracle Database SQL 1Z0-071	2Hr.	Free	-----	III Sem	Software	**Oracle**	https://mylearn.oracle.com/ou/home

CSS (Basic)	20 mins assessment	Free	Covers topics like Cascading and Inheritance, CSS Layouts, Text Styling Fundamentals, Boxing of Elements in CSS,	III Sem	Software	**Hacker Rank**	https://www.hackerrank.com/skills-verification
JavaScript (Basic)	1 hr 30 mins assessment	Free	It covers topics like, Functions, Currying, Hoisting, Scope, Inheritance, Events and Error Handling.	III Sem	Software	**Hacker Rank**	https://www.hackerrank.com/skills-verification
Problem Solving (Basic)	1 hr 30 mins assessment	Free	It covers basic topics of Data Structures (such as Arrays, Strings) and Algorithms (such as Sorting and Searching).	III Sem	Software	**Hacker Rank**	https://www.hackerrank.com/skills-verification
SQL (Basic)	30 mins assessment	Free	It includes simple queries, relationships, and aggregators.	III Sem	Software	**Hacker Rank**	https://www.hackerrank.com/skills-verification
C# (Basic)	1 hr assessment	Free	Covers topics like Exception Handling, C# Program Structure, Properties and Indexers, Types and Variables	III Sem	Software	**Hacker Rank**	https://www.hackerrank.com/skills-verification
Learn HTML from Scratch	Duration: 1 Weeks Commitment: 6 Hours	Rs. 977/-	Module 1 Introduction and Source Code, Module 2 Your First Website, Module 3 Set Up Your Free Hosting with FireFTP, Module 4 Headings, Paragraphs, Links and Images, Module 5 Inline vs Block Elements, Module 6 Iframes - Activity: Embed a Nice Relaxing, YouTube Video to Your Website, Module 7 Lists, Module 8 Tables, Entities and Forms, Module 9 Text Decoration and Comments, Module 10 Test Your Knowledge	III Sem	Software	**TCS**	https://www.tcsion.com/hub/career-creator/free-learning-courses/
Build a Fruits Slice Game (HTML, CSS and jQuery)	Duration: 1 Weeks Commitment: 4 Hours	Rs. 977/-	Module 1 Project Introduction Module 2 Page Structure and Styling Module 3 Game Logic Module 4 jQuery Code Module 5 Source Code	III Sem	Software	**TCS**	https://www.tcsion.com/hub/career-creator/free-learning-courses/

Bansal Group of Institutions
Department of Training

Certificate Courses (for CSE / IT / EC / EX) - IV Sem : Software Domain + Communication Skill

CERTIFICATION COURSES_TCS ION, HP, HCL,CISCO, SAYLOR, COURSERA ETC. (IV SEM)

Note: The courses/assessment available in different platforms(free/paid) are listed here as suggestion.Other courses(important from placement/skill point of view) can also be incorporated

NAME OF COURSES	Duration	Cost	About the Course	Semester	Category	Platform	Link
ESL002: Intermediate English as a Second Language	15 hours	Free	Improve your writing accuracy and fluency by developing your English grammar and writing skills and improving your reading and vocabulary. Unit 1, you will learn how past and present tenses change the meaning of what you read and write. Unit 2 improves your ability to read actively as you skim, scan, and read intensively. Unit 3, you will read more to collect new vocabulary and learn strategies to develop it more with synonyms, antonyms, and suffixes. Unit 4 brings these skills together as you write a narrative paragraph about something you learned in the past. By the end of ESL002, you will be able to use different tenses to describe past experiences and explain to a reader how you best learn.	IV Sem	English	Saylor	https://learn.saylor.org/?utm_source=hp&utm_medium=CTA&utm_campaign=Start
Telephone Etiquette	Duration: 1 Weeks Commitment: 2 Hours	Free	Answering an official phone call has to be done with tact. Communicating on the telephone creates an impression about both the caller and the receiver. In this course you will first learn the essential guidelines of telephone etiquette. These are points you will follow when you either make or receive a call. Next, you will learn phrases for making and receiving calls. There is a difference between formal and informal conversations and you will learn about it in this course. Next, you will learn the format for taking messages, so that the message is passed on clearly and completely. After this, you will learn to place calls on hold. Finally, you will learn about voice mail and pointers to leave effective voice mail. This course will help you learn the right way to make and take official phone calls. By purchasing or subscribing to this program, you agree to abide by the Usage Policy, Privacy Policy, Security Policies and any Terms and Conditions mentioned on this portal.	IV Sem	Comm Skill	TCS	https://www.tcsion.com/hub/career-creator/free-learning-courses/
Career Development	No more than 60 minutes	Free	In this course, you will learn how to: Create an effective, tailored resume and cover letter. Prepare for an interview and make a strong impression. Conduct a self-assessment which can be used to generate personal and professional insights.	IV Sem	Comm Skill	HP	https://www.tcsion.com/hub/career-creator/free-learning-courses/
CS107 C++ Programming	40 hours	Free	Learn the C++ computer programming language, with a focus on syntax for primitive types, control structures, vectors, strings, structs, classes, functions, file I/O, exceptions, and other programming constructs. In this course, we will learn the mechanics of editing and compiling programs in C++. We will begin with a discussion of the essential elements of C++ programming: variables, loops, expressions, functions, and the string class. Then, we will cover the basics of object-oriented programming: classes, inheritance, templates, exceptions, and file manipulation. We will then review function and class templates and the classes that perform output and input of characters to/from files. This course will also cover namespaces, exception handling, and preprocessor directives. In the last part of the course, we will learn some slightly more sophisticated programming techniques that deal with data structures such as linked lists and binary trees.	IV Sem	Software	Saylor	https://learn.saylor.org/?utm_source=hp&utm_medium=CTA&utm_campaign=Start
CS201: Elementary Data Structures	38 hours	Free	Survey basic abstract data types, their associated algorithms, and how they are implemented. Topics discussed include the structures of stacks, queues, lists, sorting and selection, searching, graphs, and hashing; performance tradeoffs of different implementations; and asymptotic analysis of running time and memory usage.	IV Sem	Software	Saylor	https://learn.saylor.org/?utm_source=hp&utm_medium=CTA&utm_campaign=Start
CS301: Computer Architecture	48 hours	Free	Explore hardware/software components, assembly language, and the functional architecture and design of computers, with a focus on topics like instruction sets, processor arithmetic and control, Von Neumann architecture, pipelining, memory management, storage, and input/output. In this course, you will study the history of modern computing technology before learning about modern computer architecture and a number of its essential features, including instruction sets, processor arithmetic and control, the Von Neumann architecture, pipelining, memory management, storage, and other input/output topics. The course will conclude with a look at the recent switch from sequential processing to parallel processing by looking at the parallel computing models and their programming implications.	IV Sem	Software	Saylor	https://learn.saylor.org/?utm_source=hp&utm_medium=CTA&utm_campaign=Start
CS302: Software Engineering	41 hours	Free	Learn how to apply an engineering approach to computer software design and development by focusing on topics like life cycle models, software requirements, specification, conceptual model design, detailed design, validation and verification, design quality assurance, software design/development environments, and project management. The course will begin with an introduction to software engineering, giving you a definition of this body of knowledge, as well as a discussion of the main methodologies of software engineering. You will then learn about the Software Development Life Cycle (SDLC), major methodologies followed by software modeling using Unified Modeling Language (UML), a standardized general-purpose modeling language used to create visual models of object-oriented software.	IV Sem	Software	Saylor	https://learn.saylor.org/?utm_source=hp&utm_medium=CTA&utm_campaign=Start
CS402: Computer Communications and Networks	42 hours	Free	Explore the hardware, software, and architectural components involved in computer communications in local area networks. Topics include the basics of computer networks, switching, routing, protocols, and security. This course is designed to give you a clear understanding of how networks, from in-home local area networks, or LANs, to the massive and global Internet, are built and how they allow us to use computers to share information and communicate with one another.	IV Sem	Software	Saylor	https://learn.saylor.org/?utm_source=hp&utm_medium=CTA&utm_campaign=Start
CS403: Introduction to Modern Database Systems	42 hours	Free	Learn about database architecture and implementation by exploring Structured Query Language (SQL), including topics like file structures and access methods; database modeling, design, and user interface; the components of database management systems; and information storage and retrieval. This course will provide a general overview of databases, introducing you to database history, modern database systems, the different models used to design a database, and Structured Query Language (SQL), which is the standard language used to access and manipulate databases. Many of the principles of database systems carry to other areas in computer science, especially operating systems. Databases are often thought of as one of the core computer science topics since many other areas in the discipline have been derived from this area.	IV Sem	Software	Saylor	https://learn.saylor.org/?utm_source=hp&utm_medium=CTA&utm_campaign=Start
CPP: Advanced Programming in C++	70 hours	Determined by Institution	Core Skills: Learn the C++ template mechanism and write your own C++ programs. Understand and use elements of the C++ STL library. Solve programming problems with STL-predefined classes and methods. Prepare for the C++ Certified Professional Programmer Certification (CPP).	IV Sem	Software	CISCO	https://www.cisco.com/site/us/en/learn/training-certifications/index.html
JavaScript Essentials 1 (JSE)	40 hours	Free*	core skills: Think algorithmically and analyze problems. Design, develop, and improve JavaScript programs. Understand a programmer's work in the software development process and the role of fundamental development tools. Know how a program is interpreted and executed in an actual local and remote computer environment. Create and develop your own programming portfolio.	IV Sem	Software	CISCO	https://www.cisco.com/site/us/en/learn/training-certifications/index.html
CLP: Advanced Programming in C	70 hours	Determined by Institution	Core Skills: Grow your knowledge of the syntax and semantics of the C language. Design and write C programs regardless of the hardware of software platform. Identify code bugs and solve complex problems using data structures and algorithms. Prepare for vendor-neutral C Certified Professional Programmer (CLP) Certification.	IV Sem	Software	CISCO	https://www.cisco.com/site/us/en/learn/training-certifications/index.html
Network Security	70 hours	Determined by Institution	Core Skills: Develop an in-depth understanding of network security. Design, implement, and support security for networked devices and data. Earn critical thinking and problem-solving skills using real equipment and Cisco Packet Tracer. Gain industry recognized skills aligned with the National Institute for Standards and Technology (NIST) Cybersecurity Framework.	IV Sem	Software	CISCO	https://www.cisco.com/site/us/en/learn/training-certifications/index.html

Build an Online Notes App (HTML, CSS, Bootstrap, PHP, MySQL and AJAX)	**Duration: 3 Weeks Commitment: 10 Hours/ Week**	**Rs. 977/-**	**Module 1 Project Introduction and Source Code, Module 2 App Skeleton and Decoration, Module 3 App Logic - Signup, Login, Remember Me, Forgot Password Module 4 Signup/Login Code Structure, Module 5 Signup/Login Code, Module 6 Remember Me, Module 7 Forgot Password Code Module 8 Notes Management: Logic and Code Structure Module 9 Load, Create, Edit and Delete Notes Module 10 Update Username, Password and Email**	**IV Sem**	**Software**	**TCS**	**https://www.tcsion.com/hub/career-creator/free-learning-courses/**
IBM: Introduction to Web Development with HTML, CSS, JavaScript	**13 hours 3 minutes**	**FREE**	**Introduction to Application Development HTML Overview CSS Overview & HTML5 Elements JavaScript Programming for Web Applications Career Opportunities and Final Project**	**IV Sem**	**Software**	**IBM**	**https://www.classcentral.com/course/introduction-to-web-development-with-html-css-jav-53747**
IBM- Machine Learning with Python	**13 hours 24 minutes**	**FREE**	**Introduction to Machine Learning In this module, you will learn about applications of Machine Learning in different fields such as health care, banking, telecommunication, and so on. You'll get a general overview of Machine Learning topics such as supervised vs unsupervised learning, and the usage of each algorithm. Also, you understand the advantage of using Python libraries for implementing Machine Learning models. Regression In this module, you will get a brief intro to regression. You learn about Linear, Non-linear, Simple and Multiple regression, and their applications. You apply all these methods on two different datasets, in the lab part. Also, you learn how to evaluate your regression model, and calculate its accuracy. Classification In this module, you will learn about classification technique. You practice with different classification algorithms, such as KNN, Decision Trees, Logistic Regression and SVM. Also, you learn about pros and cons of each method, and different classification accuracy metrics. Linear Classification Clustering In this module, you will learn about clustering specifically k-means clustering. You learn how the k-means clustering algorithm works and how to use k-means clustering for customer segmentation. Final Exam and Project In this module, you will do a project based of what you have learned so far. You will submit a report of your project for peer evaluation.**	**V Sem**	**Software**	**IBM**	**https://www.classcentral.com/course/introduction-to-web-development-with-html-css-jav-53747**
Prepare for Oracle Database SQL Certification	4 Modules 1h 27m	**Free**	Benefits to you Prepare for Oracle Database SQL Certification Additional Information Getting Started and New Releases : Exam Preparation Role : Database Developer Learn about Oracle's Products : SQL Version : 19c Language : English	**IV Sem**	**Software**	**Oracle**	**https://mylearn.oracle.com/ou/course/mysql-80-for-developers/81909**
Exam: Oracle Java Foundations (1Z0-811-A)	180 Minutes	**Free**	-	**IV Sem**	**Software**	**Oracle**	**https://mylearn.oracle.com/ou/home**
Python (Basic) Skills Certification Test	1 hr 30 mins assessment	**Free**	Covers topics like Objects and Types	**IV Sem**	**Software**	**Hacker Rank**	**https://www.hackerrank.com/skills-verification**
Frontend Developer (React)	1 hr assessment	**Free**	Demonstrate your React, CSS, Javascript Skills	**IV Sem**	**Software**	**Hacker Rank**	**https://www.hackerrank.com/skills-verification**
Java (Basic)	1 hr assessment	**Free**	Covers topics like Classes, Inheritance, Data Structures, Exception Handling	**IV Sem**	**Software**	**Hacker Rank**	**https://www.hackerrank.com/skills-verification**
Node (Basic)	1 hr assessment	**Free**	It covers topics like Package and Modules Management, Callbacks, Event Loop, Event Emitter, Buffers, Streams and File Systems	**IV Sem**	**Software**	**Hacker Rank**	**https://www.hackerrank.com/skills-verification**
Problem Solving (Intermediate)	1 hr 30 mins assessment	**Free**	It covers topics of Data Structures (such as HashMaps, Stacks and Queues) and Algorithms (such as Optimal Solutions).	**IV Sem**	**Software**	**Hacker Rank**	**https://www.hackerrank.com/skills-verification**
React (Basic)	1 hr 30 mins assessment	**Free**	React (Basic) It covers topics like Basic Routing, Rendering Elements,State Management (Internal Component State), Handling Events, ES6 and JavaScript and Form Validation.	**IV Sem**	**Software**	**Hacker Rank**	**https://www.hackerrank.com/skills-verification**
SQL (Intermediate)	35 mins assessment	**Free**	It includes complex joins, unions, and sub-queries.	**IV Sem**	**Software**	**Hacker Rank**	**https://www.hackerrank.com/skills-verification**

	ration	Cost	About the Course	Semester	Category	Platform	Link
COMM411: Public Relations	43 hrs	Free	Prepare for public relations work in small start-up businesses, international companies, political campaigns, social programs, personal development, or other outreach projects. This course will help prepare you to conduct public relations suitable for small start-up businesses, international companies, political campaigns, social programs, personal development, and other outreach projects.	V Sem	Comm Skills	Saylor	https://learn.saylor.org/?utm_source=hp&utm_medium=CTA&utm_campaign=Start
ENGL001: English Composition I	32 hours	Free	Learn how to write clear, grammatically-sound expository and persuasive prose pieces in this introductory English composition course. unit 2 will focus on academic writing. We will learn how to respond to an assignment or test question by using the "PWR-Writing" or "Power-Writing" Method (PWR: prewrite, write, revise) while learning the ins and outs of building a solid thesis and supporting that thesis with evidence. The remaining units will focus on good writing practices, from style to proper citation.	V Sem	Comm Skills	Saylor	https://learn.saylor.org/?utm_source=hp&utm_medium=CTA&utm_campaign=Start
ESL003: Upper-Intermediate English as a Second Language	12 hours	Free	Improve your reading and writing skills through active reading and academic writing. This course is designed for learners at the Upper-Intermediate level (B2). Unit 1, you will learn how sentences become clearer when tenses agree. You'll practice recognizing tenses to make your writing more coherent. Unit 2 builds your vocabulary by exploring synonyms and antonyms. Unit 3, you will develop stronger reading skills when you practice making inferences and explaining your thoughts on what you've read. Unit 4 brings all these topics together as you write a short, cohesive essay about an article. By the end of this course, you will be able to describe your opinions about the things you read.	V Sem	Comm Skills	Saylor	https://learn.saylor.org/?utm_source=hp&utm_medium=CTA&utm_campaign=Start
ESL004: Advanced English as a Second Language	12 hours	Free	Improve your academic reading and writing skills using concept mapping and critiques. This course is designed for English as a Second Language learners at the Advanced level (C1). Unit 1, you will strengthen your grammar skills by learning the basics of punctuation. You'll examine how changing punctuation can change the entire meaning of a sentence. Unit 2 connects reading to building a robust vocabulary. You'll explore how mapping a reading allows you to find new words to explain things and see different concepts. Unit 3, you'll pull everything together by drawing your own conclusions about what you've read and explaining those conclusions to your readers. By the end of ESL004, you'll be able to use punctuation, vocabulary, and mapping to write a short, cohesive essay that will prepare you for future academic work.	V Sem	Comm Skills	Saylor	https://learn.saylor.org/?utm_source=hp&utm_medium=CTA&utm_campaign=Start
PRDV002: Professional Writing	4 hrs	Free	Get the basic skills you need as an adult learner to write effective documents in the workplace. This course is designed to provide adult learners with the basic skills they need to write effective documents in the workplace. In this course, you will learn how to analyze your audience so that you can write prose that is both clear and persuasive. You will practice writing common business documents, such as emails, memos, proposals, and presentations. You will also learn how to effectively edit these documents for maximum impact.	V Sem	Comm Skills	Saylor	**https://learn.saylor.org/?utm_source=hp&utm_medium=CTA&utm_campaign=Start**
PRDV006: Spreadsheets II: Formatting and Functions	6 hrs	Free	Learn how to format spreadsheets with formulas and functions, build multi-page spreadsheets, chart data, create tables with database features, and create pivot tables. This course is intended for students who have a basic understanding of spreadsheets and are ready to delve into formatting, formulas, functions, multi-page spreadsheets, charting data, creating tables with database features, and pivot tables. This course builds on PRDV004: Spreadsheets, so be sure to review basic formatting commands of Excel and Google Sheets	V Sem	Comm Skills	Saylor	https://learn.saylor.org/?utm_source=hp&utm_medium=CTA&utm_campaign=Start
PRDV008: Preparing and Delivering Presentations	22 hrs	Free	Learn how to prepare and deliver a presentation on any topic that demonstrates an organizational pattern, with supporting materials and appropriate references that support or establish credibility and authority. Explore how to use proper tone, volume, rate, and modulation of voice, body language, eye contact, and other forms of nonverbal communication. Discover how to present online, and strategies for managing public speaking anxiety and planning for the unexpected.	V Sem	Comm Skills	Saylor	Saylor Academy
PRDV104: Professional Etiquette	06 hours	Free	Learn about typical workplace etiquette protocols, communication standards, and cultural awareness strategies in order to navigate common workplace obstacles as smoothly as possible. We begin by introducing professional manners, respect, and common courtesy. Then, we discuss effective communication in terms of verbal, nonverbal, and virtual communication techniques. Finally, we investigate diversity and how we should respond to and respect cultural differences to create a friendlier and more inclusive workplace.	V Sem	Comm Skills	Saylor	https://learn.saylor.org/?utm_source=hp&utm_medium=CTA&utm_campaign=Start
PRDV228: Decision-Making	4 hours	Free	Learn about the decision-making process, including basic decision types, tools, methods, and the insights that help you realize how decision-making involves both logic and emotion. In this course, we will look at the decision-making process, paying close attention to the basic decision types, tools, methods, and insights to help you quickly learn how it involves both logic and emotion. When you complete the course, you will have practical tools to quickly determine the type of decision you are trying to make, the available tools and methods you may use, and the way to effectively engage your teams in the process.	V Sem	Comm Skills	Saylor	https://learn.saylor.org/?utm_source=hp&utm_medium=CTA&utm_campaign=Start
Presenting Data	No more than 60 minutes	Free	In this course, you will learn how to: Understand how to use different charts and graphs to help people understand and value your business. Decide what type of chart to use to present your data effectively. Use spreadsheet software to create charts and graphs.	V Sem	Comm Skills	HP	https://www.life-global.org/allcourses?page=1
All Courses Marketing Customer Relationship Management	No more than 60 minutes	Free	In this course, you will learn how to: Set up a CRM tool for your business. Use the CRM tool to track information that will help you build your business.	V Sem	Comm Skills	HP	https://www.life-global.org/allcourses?page=1
Social Media Marketing	No more than 60 minutes	Free	In this course, you will learn how to: Use different social media platforms to promote your business. Create a Facebook advertisement targeted to your customers.	V Sem	Comm Skills	HP	https://www.life-global.org/allcourses?page=1
Effective Leadership	No more than 60 minutes	Free	In this course, you will learn how to: Understand effective leadership and its importance. Cultivate your own leadership style and skills. Leverage leadership strategies to build, manage, and lead strong teams. Facilitate and navigate change. Learn how ethics play a role in effective leadership.	V Sem	Comm Skills	HP	https://www.life-global.org/allcourses?page=1
Data Science & Analytics	No more than 60 minutes	Free	In this course, you will: Examine the benefits and challenges of data-driven approaches for businesses. Consider strategies for leveraging data to gain a competitive advantage across various industries. Recognize the essential skills and tools required to pursue a career in data science and analytics. Define machine learning and artificial intelligence and describe their role in reshaping our approach to data and technology. Explore the relationship between data and technology and identify the potential ethical and societal implications they each have on the Digital Era. Develop strategies to mitigate and address cybersecurity threats in data applications. This course was jointly developed by HP Inc. Digital & Transformation Organization and HP LIFE.	V Sem	Comm Skills	HP	https://www.life-global.org/allcourses?page=1
CS105 Introduction to Python	36 hours	Free	Learn fundamental programming concepts using the Python 3 programming language, a high-level interpreted language that is easy to read write, with powerful libraries that provide additional functionality. This course is an introduction to fundamental programming concepts by way of the Python 3 programming language. Python 3 is a high-level interpreted language that has many benefits, including easy-to-read and easy-to-write syntax and powerful libraries that provide additional functionality. Even though Python 3 is a great programming language for beginners, it is also used extensively for practical applications in engineering and data science. This course is intended for people with no or very little prior programming experience. It covers a range of topics, such as data types, control flow, functions, file operations, and object-oriented programming. When you finish this course, you will be able to create Python programs for a variety of applications.	V Sem	Software	Saylor	https://learn.saylor.org/?utm_source=hp&utm_medium=CTA&utm_campaign=Start

BANSAL GROUP OF INSTITUTES BHOPAL|INDORE|MANDIDEEP	ıration	Cost	About the Course	Semester	Category	Platform	Link
COMM411: Public Relations	43 hrs	Free	Prepare for public relations work in small start-up businesses, international companies, political campaigns, social programs, personal development, or other outreach projects. This course will help prepare you to conduct public relations suitable for small start-up businesses, international companies, political campaigns, social programs, personal development, and other outreach projects.	V Sem	Comm Skills	Saylor	https://learn.saylor.org/?utm_source=hp&utm_medium=CTA&utm_campaign=Start
Python for Data Science	67 hours	Free	Learn data science using the Python programming language by looking at data processing, data analysis, visualization, data mining, and statistical models. By the end of this course, you will be able to implement Python code for these data science topics. This course attempts to strike a balance between presenting the vast set of methods within the field of data science and Python programming techniques for implementing them. Problem-solving and programming implementation will be emphasized throughout the course. All techniques presented will be introduced using real-world programming examples. A major goal of the course is to ensure that when you finish the course, you will have the programming and conceptual expertise you need to join the field of data science.	V Sem	Software	Saylor	https://learn.saylor.org/?utm_source=hp&utm_medium=CTA&utm_campaign=Start
IoT Fundamentals: Big Data & Analytics	50 hours	Determined by Institution	Core Skills: Use Python to create code that reads data from sensors and stores it in a SQL database. Visualize, clean, manipulate and integrate data sets. Learn fundamental principles of Big Data platforms like Hadoop. Use storytelling to present insights gained from extracted data.	V Sem	Software	CISCO	https://www.cisco.com/site/us/en/learn/training-certifications/index.html
IoT Fundamentals: Hackathon Playbook	30 hours	Determined by Institution	Core Skills: Identify a problem, create a design, then plan for an IoT solution. Develop a prototype of an IoT solution. Test and validate your prototype. Present IoT concept and solution to experts.	V Sem	Software	CISCO	https://www.cisco.com/site/us/en/learn/training-certifications/index.html
PCAP: Programming Essentials in Python	75 hours	Free*	core skills: Think algorithmically – how to analyze a problem and translate it for a computer to process. Design, develop, and improve multi-module computer programs. Analyze and model real-life problems in Object-Oriented Programming categories. Understand a programmer's work in the software development process. Learn how a program is executed in a computer environment. Gain skills to create and develop your own programming portfolio.	V Sem	Software	CISCO	https://www.cisco.com/site/us/en/learn/training-certifications/index.html
CCNA: Introduction to Networks	70 hours	Determined by Institution	Core Skills: Build simple LANs, perform basic configurations for routers and switches, and implement IPv4 and IPv6 addressing schemes. Configure routers, switches, and end devices to provide access to local and remote network resources and to enable end-to-end connectivity between remote devices. Develop critical thinking and problem-solving skills using real equipment and Cisco Packet Tracer. Configure and troubleshoot connectivity a small network using security best practices.	V Sem	Software	CISCO	https://www.cisco.com/site/us/en/learn/training-certifications/index.html
CCNA: Switching, Routing, and Wireless Essentials	70 hours	Determined by Institution	Core Skills: Work with routers, switches and wireless devices to configure and troubleshoot VLANs, Wireless LANs and Inter-VLAN routing. Configure and troubleshoot redundancy on a switched network using STP and EtherChannel. Develop critical thinking and problem-solving skills using real equipment and Cisco Packet Tracer. Explain how to support available and reliable networks using dynamic addressing and first-hop redundancy protocols.	V Sem	Software	CISCO	https://www.cisco.com/site/us/en/learn/training-certifications/index.html
DevNet Associate	70 hours	Determined by Institution	Core Skills: Practice software development skills, including Python, GIT, and common data formats (XML, JSON, and YAML). Become familiar with infrastructure automation using code, DevOps methodology, and microservices. Deploy applications as containers and use Continuous Integration/Continuous Deployment (CI/CD) pipelines. Develop critical thinking and problem-solving skills using Cisco Packet Tracer and virtual machines.	V Sem	Software	CISCO	https://www.cisco.com/site/us/en/learn/training-certifications/index.html
IoT Fundamentals: IoT Security	50 hours	Determined by Institution	Core Skills: Conduct end-to-end security assessments of IoT systems to demonstrate vulnerabilities. Gain hands-on experience with IoT prototypes using a Raspberry Pi. Recommend threat mitigation measures to minimize the risk in IoT solutions and networks. Become proficient using real-world penetration and vulnerability testing tools such as Kali Linux.	V Sem	Software	CISCO	https://www.cisco.com/site/us/en/learn/training-certifications/index.html
TCS MasterCraftTM DevPlus Overview	Duration: 1 Weeks Commitment: 2 Hours	free	TCS MasterCraftTM DevPlus is a scalable, Agile and DevOps product that helps learners tailor their workflows to track and manage Scrum, Kanban or any other Agile frameworks; perform continuous testing and automate release management. This interactive course consists of videos, self-paced learning modules, knowledge check questions and a game-based quiz at the end of the course. The course design is compatible with all devices. On completion of the course, the learners will have knowledge about the structure of TCS MasterCraft™ DevPlus, its features and the operations that can be performed. The learners will also gain an understanding of the fundamental components of TCS MasterCraft™ DevPlus with a focus on the Agile Project Management, Application Release Automation, Quality Management and Platform Capabilities. They would be able to confidently explore the features of the tool and apply it to their projects, or promote the features of this product to their customers.	V Sem	Software	TCS	**https://www.tcsion.com/hub/career-creator/free-learning-courses/**
Learn PHP and MySQL from Scratch	Duration: 5 Weeks Commitment: 10 Hours/Week	Rs. 977/-	Module 1 Introduction to PHP and Source Code, Module 2 PHP Variables and Data Types, Module 3 String Functions, Module 4 Conditional and Looping, Module 5 Functions, Module 6 GET and POST, Module 7 Array Functions, Module 8 PHP Emails and Filtering Inputs, Module 9 Activity: Create a Responsive Contact Form Using PHP and Bootstrap, Module 10 Date and Time, Module 11 Files and File Handling, Module 12 Uploading Files, Module 13 Cookies, Module 14 Error Handling, Module 15 Sessions, Module 16 Test Your Knowledge, Module 17 Introduction to MySQL and Source Code, Module 18 PHP My Admin and Remote MySQL, Module 19 Use PHP and MySQL to Create/Populate Databases and Tables, Module 20 Activity: Populate Database Once the User Submits a Form, Module 21 Activity: Populate an HTML Table Using Database Data, Module 22 Update/Delete Data, Module 23 Test Your Knowledge	V Sem	Software	TCS	https://www.tcsion.com/hub/career-creator/free-learning-courses/
Introduction to Artificial Intelligence with Java	"Duration: 1 Weeks Commitment: 2-3 Hours"	Rs. 1115/-	Module 1 Introduction to Artificial Intelligence and Java Module 2 Exploring Search Module 3 AI Games and Rule-Based System Module 4 Interfacing with Weka Module 5 Handling Attributes Module 6 Supervised Learning Module 7 Semi-Supervised and Unsupervised Learning	V Sem	Software	TCS	https://www.tcsion.com/hub/career-creator/free-learning-courses/
Python for Data Science, AI & Development	25 hours (approximately)	FREE	Python Basic Python Data Structure Python Programming Fundamental Working with Data in Python API and Data Collection	V Sem	Software	Google	
Programming in Python	44 hours (approximately)	FREE	Get started with the Python programming language and associated foundational concepts. Learn basic Python syntax, to use control flow and loops and work with functions and data structures. You will also learn how to recognise possible errors, their causes and how to handle them. Learn about the paradigms of procedural programming and the associated logical concepts. You'll explore functional and object-oriented programming, and get an introduction to algorithms. Supercharge your coding environment with popular modules libraries and tools for Python. You'll also learn about the different types of testing and how to write a test. Practice and reflect on the skills you learned in this course.	V Sem	Software	Google	https://groen_in/certificates/
IBM- Python for Data Science, AI & Development	1 day 1 hour 3 minutes	FREE	Python Basics This module teaches the basics of Python and begins by exploring some of the different data types such as integers, real numbers, and strings. Continue with the module and learn how to use expressions in mathematical operations, store values in variables, and the many different ways to manipulate strings. Python Data Structures This module begins a journey into Python data structures by explaining the use of lists and tuples and how they are able to store collections of data in a single variable. Next learn about dictionaries and how they function by storing data in pairs of keys and values, and end with Python sets to learn how this type of collection can appear in any order and will only contain unique elements. Python Programming Fundamentals This module discusses Python fundamentals and begins with the concepts of conditions and branching. Continue through the module and learn how to implement loops to iterate over sequences, create functions to perform a specific task, perform exception handling to catch errors, and how classes are needed to create objects. Working with Data in Python This module explains the basics of working with data in Python and begins the path with learning how to read and write files. Continue the module and uncover the best Python libraries that will aid in data manipulation and mathematical operations. APIs and Data Collection This module delves into the unique ways to collect data by the use of APIs and web scraping. It further explores data collection by explaining how to read and collect data when dealing with different file formats.	V Sem	Software	IBM	Than you tell me what is your answer

	ıration	Cost	About the Course	Semester	Category	Platform	Link
COMM411: Public Relations	43 hrs	Free	Prepare for public relations work in small start-up businesses, international companies, political campaigns, social programs, personal development, or other outreach projects. This course will help prepare you to conduct public relations suitable for small start-up businesses, international companies, political campaigns, social programs, personal development, and other outreach projects.	V Sem	Comm Skills	Saylor	https://learn.saylor.org/?utm_source=hp&utm_medium=CTA&utm_campaign=Start
Python Basics for Data Science	3 weeks, 4-10 hours a week	Paid	Module 1 - Python Basics : Your first program, Types Expressions and Variables String Operations Module 2 - Python Data Structures : Lists and Tuples, Sets Dictionaries Module 3 - Python Programming Fundamentals : Conditions and Branching, Loops, Functions Objects and Classes Module 4 - Working with Data in Python : Reading files with open, Writing files with open, Loading data with Pandas Working with and Saving data with Pandas Module 5 - Working with Numpy Arrays : Numpy 1d Arrays Numpy 2d Arrays	V Sem	Software	IBM	https://www.classcentral.com/course/introduction-to-web-development-with-html-css-jav-52747
Java		FREE	Program in Java using Java SE 8 and Java SE 11, the long-term support releases, and most widely used versions of Java SE. Program using Java EE 7, which is the platform for enterprise Java applications. Use the module system to design applications with explicit dependencies and encapsulation at the JAR level, ensuring more reliable configuration, improved security, and enhanced performance. Make your code more readable and succinct using convenience methods. Test code snippets and APIs using JShell. Use Java Persistence and Java Transaction APIs, create a flexible component model using EJB and CDI technology, provide Timer, Concurrency and Batch services, and create SOAP and REST web services. Develop web-based interfaces using Servlets, Java Server Pages, and JavaServer Faces. Implement best practices for writing Java code. Leverage product enhancements soon after each product release. Take advantage of all training in the learning subscription in order to learn and get certified on Java SE 8, Java SE 11, and Java EE 7. Learn to implement WebLogic Server, SOA Cloud, and WebCenter Portal cloud services in creating applications on Oracle Cloud.	V Sem	Software	Oracle	https://mylearn.oracle.com/ou/course/mysql-80-for-developers/81909
Software Engineer	1 hr 15 mins assessment	FREE	Demonstrate your Problem Solving, SQL, REST API Skills	V Sem	Software	Hacker Rank	https://www.hackerrank.com/skills-verification
Software Engineer Intern	1 hr assessment	FREE	Demonstrate your Problem Solving, SQL Skills	V Sem	Software	Hacker Rank	https://www.hackerrank.com/skills-verification
JavaScript (Intermediate)	1 hr 30 mins assessment	FREE	It covers topics like Design Patterns, Memory management, concurrency model, and event loops, among others.	V Sem	Software	Hacker Rank	https://www.hackerrank.com/skills-verification
SQL (Advanced)	1 hr assessment	FREE	It covers topics like query optimization, data modeling, Indexing, window functions, and pivots in SQL.	V Sem	Software	Hacker Rank	https://www.hackerrank.com/skills-verification

Bansal Group of Institutions
Department of Training

Certificate Courses (for CSE / IT / EC / EX) - VI Sem : Software Domain + Communication Skill

CERTIFICATION COURSES_TCS ION, HP, HCL,CISCO, SAYLOR, COURSERA ETC. (VI SEM)

Note: The courses/assessment available in different platforms(free/paid) are listed here as suggestion.Other courses(important from placement/skill point of view) can also be incorporated

NAME OF COURSES	Duration	Cost	About the Course	Semester	Category	Platform	Link
ENGL002: English Composition II	101 hrs	Free	This course will reinforce the concepts you practiced in ENGL001 by introducing you to basic research concepts and techniques Unit 1, you will select a topic that intrigues you, conduct preliminary research to focus your topic, and develop a thesis statement Unit 2, you will learn strategies for conducting your research and taking careful notes. Unit 3, you will learn how to evaluate and understand the sources you located in the previous units Unit 4, you will develop your argument and create a detailed outline for your research paper Unit 5 focuses on how to use style standards and citation methodology correctly.	VI Sem	Comm Skill	Saylor	https://learn.saylor.org/?utm_source=hp&utm_medium=CTA&utm_campaign=Start
ENGL210: Technical Writing	86 hrs	Free	to improve your ability to write clear, comprehensible examples of technical writing, most subunits include short writing activities that will give you hands-on experience in many different writing tasks	VI Sem	Comm Skill	Saylor	https://learn.saylor.org/?utm_source=hp&utm_medium=CTA&utm_campaign=Start
ESL005: Business-Proficient English as a Second Language	12 hours	Free	Recognize and employ business writing styles in Standard American English in this English as a Second Language course designed for learners at the Proficient level (C2). Unit 1, you will examine Standard American English in business contexts. You'll consider how writers use direct and indirect communication and what terminology is applied. Unit 2 looks at the different genres of writing in the business world. You'll recognize the differences between emails, memos, and letters. You'll see why each is used and examine when each type of writing is used. Finally, you'll write a business memo based on a real-life case and use your language skills to make your ideas clear to your readers. By the end of this course, you'll be able to recognize different business writing styles and know when to use them.	VI Sem	Comm Skill	Saylor	https://learn.saylor.org/?utm_source=hp&utm_medium=CTA&utm_campaign=Start
PRDV007: Spreadsheets III: Presenting Data	5 hrs	Free	Present data using charts, plots, graphics, and tables. Learn how to create and format charts and scatterplots, and how to add graphics to communicate and organize information better. This course focuses on presenting data using charts, plots, graphics, and tables. We will learn how to create and format the appropriate chart for a given set of data. We will also learn how to create scatterplots for correlated data and fit those plots to an equation. We will add appropriate graphics to worksheets that will help us communicate better with users. Finally, we will use the database functions of tables to organize information better	VI Sem	Comm Skill	Saylor	https://learn.saylor.org/?utm_source=hp&utm_medium=CTA&utm_campaign=Start
PRDV009: Writing Grant Proposals	12 hrs	Free	Explore how to write a coherent, persuasive grant proposal to obtain funding for a project or initiative by focusing on topics like the basics of nonprofit organizations, how to interpret RFPs, the elements of a grant proposal, and how to find funding sources and build relationships with funders.	VI Sem	Comm Skill	Saylor	https://learn.saylor.org/?utm_source=hp&utm_medium=CTA&utm_campaign=Start
PRDV200: Communicating with Data	8 hours	Free	Get a competitive edge in the marketplace by understanding data visualization, performing quantitative and qualitative data analysis, crafting data visualizations, and presenting results. This course is designed to help industry professionals gain a competitive edge in the marketplace by understanding data visualization. We will analyze quantitative and qualitative data, craft visualizations, and present results. We will also learn how to manage data, perform analytics with a set of charts, and organize information into visualizations to tell a story. By developing your knowledge of creating and presenting data-driven insights, you can solve applied problems in better ways.	VI Sem	Comm Skill	Saylor	https://learn.saylor.org/?utm_source=hp&utm_medium=CTA&utm_campaign=Start
PRDV212: Interpersonal Communication	14 hours	Free	Investigate interpersonal communication in formal and informal settings by building and managing personal and professional relationships through effective communication This course will define interpersonal communication and apply its basic elements to everyday situations. You will investigate interpersonal communication in formal and informal settings and discuss conflict management	VI Sem	Comm Skill	Saylor	https://learn.saylor.org/?utm_source=hp&utm_medium=CTA&utm_campaign=Start
Business Email	No more than 60 minutes	Free	In this course, you will learn how to: Recognize the dos and don'ts of business email. Address an email. Add an attachment to an email. Create an email signature.	VI Sem	Comm Skill	HP	https://www.life-global.org/allcourses?page=1
Business Communications	No more than 60 minutes	Free	In this course, you will learn how to: Identify the characteristics of effective business communications. Identify success factors for effective communication. Identify and analyze your target audience. Use communication mediums to effectively convey messages.	VI Sem	Comm Skill	HP	https://www.life-global.org/allcourses?page=1
CS406: Information Security	46 hours	Free	Learn the principles of information security to protect the confidentiality, integrity, and availability of information. Discuss the modes of threats and attacks on information systems, threat mitigation, cryptography, user identification and authentication, access control, privacy laws, and more. This course covers information security principles, an area of study that engages in protecting the confidentiality, integrity, and availability of information. Information security continues to grow with advancements in technology – as technology advances, so do threats, attacks, and our efforts to mitigate them. In this course, we discuss the modes of threats and attacks on information systems. We also discuss an important area of threat mitigation that saw rapid development in the twentieth century: cryptography	VI Sem	Software	Saylor	https://learn.saylor.org/?utm_source=hp&utm_medium=CTA&utm_campaign=Start
NDG Linux Unhatched	08 hours	Free*	Core Skills: Basic installation and configuration of Linux software. Understand the basics of the Linux Command Line Interface (CLI). Interact with the Linux virtual machine. Determine if Linux is for you or not.	VI Sem	Software	CISCO	https://www.cisco.com/site/us/en/learn/training-certifications/index.html
NDG Linux Essentials	70 hours	Free*	Core Skills: Acquire open source concepts and progressively master Linux commands. Understand how Linux is used and the basics of its command line. Become skilled at using the Linux virtual machine for experiments. Prepare for LPI Linux Essentials Professional Development Certificate (PDC).	VI Sem	Software	CISCO	https://www.cisco.com/site/us/en/learn/training-certifications/index.html
Python for Everybody Specialization	2 months at 10 hours a week	Free	Programming for Everybody (Getting Started with Python) Python Data Structures Using Python to Access Web Data Using Databases with Python Capstone: Retrieving, Processing, and Visualizing Data with Python	VI Sem	Software	Google	https://grow.google/intl/en_in/certificates/

Machine Learning with Python	3 hours	Free	**Module 1 - Supervised vs Unsupervised Learning : Machine Learning vs Statistical Modelling, Supervised vs Unsupervised Learning, Supervised Learning Classification Unsupervised Learning Module 2 - Supervised Learning I : K-Nearest Neighbors, Decision Trees, Random Forests, Reliability of Random Forests, Advantages & Disadvantages of Decision Trees Module 3 - Supervised Learning II : Regression Algorithms, Model Evaluation, Model Evaluation: Overfitting & Underfitting, Understanding Different Evaluation Models Module 4 - Unsupervised Learning : K-Means Clustering plus Advantages & Disadvantages, Hierarchical Clustering plus Advantages & Disadvantages, Measuring the Distances Between, Clusters - Single Linkage Clustering, Measuring the Distances Between Clusters - Algorithms for Hierarchy Clustering, Density-Based Clustering Module 5 - Dimensionality Reduction & Collaborative Filtering : Dimensionality Reduction: Feature Extraction & Selection, Collaborative Filtering & Its Challenges**	**VI Sem**	**Software**	**IBM**	https://www.classcentral.com/course/introduction-to-web-development-with-html-css-jav-53747
Developing Back-End Apps with Node.js and Express	13 hours 9 minutes	Free	**Introduction to Server-Side JavaScript You may already be familiar with JavaScript, but did you know there is a server-side JavaScript framework? Discover how Node.js is used on the back end to develop rich, dynamic web applications by importing external modules and using node package manager (NPM). Get hands-on experience using Node.js to build your own simple web server. Asynchronous I/O with callback programming How can you optimize processor time with Node.js? Discover how Node.js operates in a non-blocking manner and how to handle the result from a network call. Explore the advantages and challenges associated with using anonymous callback functions. Use Node.js to parse JavaScript Object Notation (JSON) data from a Hypertext Transfer Protocol (HTTP) message. Express Web Application Framework When you use Node.js to build a hypertext transfer protocol (HTTP) server, you will discover that the default Node.js framework is low-level. You can extend its features with third-party packages such as Express. Get hands-on experience with the Express web application framework by creating a web server that listens to incoming requests. You also learn how to implement authorization with Express. Use routing, middleware, and template rendering to extend the Node.js framework and handle dynamic content. Final Project In this module, you will complete two related projects: a practice project and a final project. The practice project walks you through code that performs CRUD operations for given user details on an Express server using session and JWT authentication and then tests the code using Postman. For the final project, you assume the role of a back-end developer working on a server-side application that stores, retrieves, and manages book ratings and reviews. You will submit screenshots of your application, and then your peers will review the project based on a rubric provided. You will also be expected to review two of your peer's projects.**	**VI Sem**	**Software**	**IBM**	https://www.classcentral.com/course/introduction-to-web-development-with-html-css-jav-53747
IBM Front-End Developer		Free	**Course 1: Getting Started with Front-End and Web Development Course 2: Introduction to Software Engineering Course 3: Designing User Interfaces and Experiences (UI/UX) Course 4: Introduction to HTML, CSS, & JavaScript Course 5: Developing Websites and Front-Ends with Bootstrap Course 6: Getting Started with Git and GitHub Course 7: Developing Front-End Apps with React Course 8: Intermediate Web and Front-End Development Course 9: Get Started with Cloud Native, DevOps, Agile, and NoSQL Course 10: Front-End Development Capstone Project Course 11: Software Developer Career Guide and Interview Preparation**	**VI Sem**	**Software**	**IBM**	https://www.classcentral.com/course/introduction-to-web-development-with-html-css-jav-53747
IBM Full-Stack JavaScript Developer		Free	**Course 1: Introduction to Software Engineering Course 2: Introduction to HTML, CSS, & JavaScript Course 3: Getting Started with Git and GitHub Course 4: JavaScript Programming Essentials Course 5: Developing Front-End Apps with React Course 6: Developing Back-End Apps with Node.js and Express Course 7: Get Started with Cloud Native, DevOps, Agile, and NoSQL Course 8: Introduction to Containers w/ Docker, Kubernetes & OpenShift Course 9: Application Development using Microservices and Serverless Course 10: Node.js & MongoDB: Developing Back-end Database Applications Course 11: JavaScript Full Stack Capstone Project Course 12: Software Developer Career Guide and Interview Preparation**	**VI Sem**	**Software**	**IBM**	https://www.classcentral.com/course/introduction-to-web-development-with-html-css-jav-53747
IBM Full Stack Software Developer	17 weeks, 10 hours a week	Free	**Course 1: Introduction to Cloud Computing Course 2: Introduction to Web Development with HTML, CSS, JavaScript Course 3: Getting Started with Git and GitHub Course 4: Developing Front-End Apps with React Course 5: Developing Back-End Apps with Node.js and Express Course 6: Python for Data Science, AI & Development Course 7: Developing AI Applications with Python and Flask Course 8: Django Application Development with SQL and Databases Course 9: Introduction to Containers w/ Docker, Kubernetes & OpenShift Course 10: Application Development using Microservices and Serverless Course 11: Full Stack Application Development Capstone Project Course 12: Full Stack Software Developer Assessment**	**VI Sem**	**Software**	**IBM**	https://www.classcentral.com/course/introduction-to-web-development-with-html-css-jav-53747
MySQL 8.0 for Developers	14h 20m	Free	**The MySQL for Developers training course teaches developers you how to develop console and web applications using MySQL with your choice of the PHP, Java, or Python programming languages. Expert Oracle University instructors will guide you through realistic hands-on activities in which you will learn how to use Connectors to access MySQL databases, query the database effectively, present data in different formats, and how to use MySQL's support for JSON and spatial data. Learn to: Program with Connectors Write console and web applications Optimize query performance Write stored routines and triggers Store, access, and analyze geospatial data Use X DevAPI to access JSON data in document store Benefits to you This course will teach you how to write applications that maximize your investment in MySQL. You will learn best practice techniques for writing programs that store and retrieve MySQL data and present it to users in a way that allows them to gain insight into that data. Additional Information Version : 8.0 Role : Database Administrator Language : English**	**VI Sem**	**Software**	**Oracle**	https://mylearn.oracle.com/ou/course/mysql-80-for-developers/81909
Oracle Database 19c: SQL Workshop	22 Modules 14h 41m	Free	**Benefits to you Learn how to: Retrieve Data using the SQL SELECT Statement Restrict and Sort Data Customize Output Using Single-Row Functions Use Conversion Functions and Conditional Expressions Report Aggregated Data Using the Group Functions Display Data from Multiple Tables Using Joins Manage Tables by Using DML Statements**	**VI Sem**	**Software**	**Oracle**	https://mylearn.oracle.com/ou/course/mysql-80-for-developers/81909
Angular (Basic) Certification	1 hr assessment	Free	**It covers topics like MVC Frameworks, Components (Angular, Dynamic, Styling), TypeScript, Two Way Binding and Form Validation**	**VI Sem**	**Software**	**Hacker Rank**	https://www.hackerrank.com/skills-verification
Node.js (Intermediate)	1 hr assessment	Free	**Understand the concepts behind an Event-Driven Architecture and Concurrency**	**VI Sem**	**Software**	**Hacker Rank**	https://www.hackerrank.com/skills-verification
R (Basic)	1 hr 30 mins assessment	Free	**This competency area includes fundamentals of the R programming language, understanding Data Frames, Packages, and Data Reshaping, using Data interfaces, among others.**	**VI Sem**	**Software**	**Hacker Rank**	https://www.hackerrank.com/skills-verification
Rest API (Intermediate)	1 hr 30 mins assessment	Free	**It covers topics like getting data from an API and process using parameters or paging.**	**VI Sem**	**Software**	**Hacker Rank**	https://www.hackerrank.com/skills-verification

TCS Assessments (for CSE / IT / EC / EX) - VI Sem : Software Domain

TCS iON - PAID ASSESSEMENT

https://www.tcsion.com

Note: The courses/assessment available in different platforms(free/paid) are listed here as suggestion.Other courses(important from placement/skill point of view) can also be incorporated

SNo	Test Name	Next Test Date	Upcoming Test Date	About the Test	How Can Apply	Part A - Test of knowledge	Syllabus	Amount/ Fee	Year/Semester, (can be suitable for)
1	**TCS iON NPT - JavaScript Developer**	**10th October, 2024**	6th November, 2024 21st December, 2024 (Applications are Open)	About the test TCS iON National Proficiency Test (TCS iON NPT) is a nation-wide assessment for jobseekers and working professionals who want to certify their knowledge and skills in JavaScript Development. All TCS iON NPT certifications are valid for 2 years. Candidates can appear for the test ONLY In-Centre at TCS iON Authorised Exam Centres. Certification tests are conducted every 2-4 weeks. However, candidates can purchase the same TCS iON NPT Variants after 90 days of their last purchase. Candidates have the opportunity to appear for multiple attempts to improve their TCS iON NPT Score on repurchases. Candidates who score 75% and above will be awarded a TCS iON NPT - JavaScript Developer Expert Certificate and those who score between 50% and 74% will be awarded a TCS iON NPT - JavaScript Developer Pro Certificate. For the rest, only the scores will be notified. Candidates with TCS iON NPT Certification can apply for jobs on the TCS iON Job Listing Portal, as well as on other job portals using this certificate.	Freshers and professionals with up to 2 years of work experience Students who are in their pre-final or final year, pursuing any degree Anyone aspiring for job roles related to JavaScript Development in top corporates	Variables, Types, and Collections: JavaScript, knowledge of JS variables, all new features in ES6 Objects, Functions, and Classes: Object-Oriented Programming and its principles, Functions and their types and more Browser and Events: Working of Web Browser, Integrating JavaScript to web browser, DOM Parsing and Event Handling and more Debugging and Error Handling: Errors, and Exception Handling Asynchronous Programming and Server-side JavaScript: JavaScript, AJAX, knowledge of Server-side JavaScript Testing: Testing techniques, their tools and test scenarios Data Storage and Security: Client-side data storage	https://www.tcsion.com/per/g01/pub/1016/iDH/instance/1/product/microsite_data/IONNPTJSD2232023/microsite_1715834844520/pdf/View%20Syllabus.pdf?version=1854575300 . . . Questions - 50 Marks - 50 Time - 60 Mins	**Rs. 799/-**	**VI Sem**
3	**TCS iON NPT - MEAN Stack Developer**	**10th October, 2024**	6th November, 2024 21st December, 2024 (Applications are Open)	About the test TCS iON National Proficiency Test (TCS iON NPT) is a nation-wide assessment for jobseekers and working professionals who want to test their knowledge and skills in MEAN Stack Development. All TCS iON NPT certifications are valid for 2 years. Candidates can appear for the test ONLY In-Centre at TCS iON Authorised Exam Centres. Certification tests are conducted every 2-4 weeks. However, candidates can purchase the same TCS iON NPT Variants after 90 days of their last purchase. Candidates have the opportunity to appear for multiple attempts to improve their TCS iON NPT Score on repurchases. Candidates who score 75% and above will be awarded a TCS iON NPT - MEAN Stack Developer Expert Certificate and those who score between 50% and 74% will be awarded a TCS iON NPT - MEAN Stack Developer Pro Certificate. For the rest, only the scores will be notified. Candidates with TCS iON NPT Certification can apply for jobs on the TCS iON Job Listing Portal, as well as on other job portals using this certificate.	Freshers and professionals with up to 2 years of work experience Students who are in their pre-final or final year, pursuing any degree Anyone aspiring for job roles related to MEAN Stack Development in top corporates	Basics, Installation and Setup: MEAN Stack Development Basics, Installation of various tools and configuring them, setting up the development environment Building User Interfaces with Angular: Front-end operations, AngularJS basics and Designing the UI with Angular Building Server-side with Node.js: Server-side Scripting, basics of Node.js, Required Modules and Components, NPM, HTTP Servers and more Building APIs with Express: Server-side operations, APIs, Express JS Managing Date with MongoDB: Database, Database Servers, NOSQL Database, MongoDB Database Putting it all together: Integration of front-end, back-end and APIs and building a high-end web application **Questions - 50, Marks - 50, Time - 60 Mins**	https://www.tcsion.com/per/g01/pub/1016/iDH/instance/1/product/microsite_data/IONNPTMSD233202/microsite_1715754888216/pdf/View%20Syllabus.pdf?version=1854575375	**Rs. 799/-**	**VI Sem**
4	**TCS iON NPT - Python Developer**	**10th October, 2024**	6th November, 2024 21st December, 2024 (Applications are Open)	About the test TCS iON National Proficiency Test (TCS iON NPT) is a nation-wide assessment for jobseekers and working professionals who want to test their knowledge and skills in Python Programming. All TCS iON NPT certifications are valid for 2 years. Candidates can appear for the test ONLY In-Centre at TCS iON Authorised Exam Centres. Certification tests are conducted every 2-4 weeks. However, candidates can purchase the same TCS iON NPT Variants after 90 days of their last purchase. Candidates have the opportunity to appear for multiple attempts to improve their TCS iON NPT Score on repurchases. Candidates who score 75% and above will be awarded a TCS iON NPT - Python Developer Expert Certificate and those who score between 50% and 74% will be awarded a TCS iON NPT - Python Developer Pro Certificate. For the rest, only the scores will be notified. Candidates with TCS iON NPT Certification can apply for jobs on the TCS iON Job Listing Portal, as well as on other job portals using this certificate.	Freshers and professionals with up to 2 years of work experience Students who are in their pre-final or final year, pursuing any degree Anyone aspiring for job roles in Python Programming in top corporates	Statistics: Basics of Statistics Core Python: Core Python with Object-oriented programming NumPy: NumPy for N-dimensional arrays Data Munging: Data munging in Python using Pandas EDA: Exploratory Data Analysis in Python using Pandas Matplotlib: Visualisation using Matplotlib	https://www.tcsion.com/per/g01/pub/1016/iDH/instance/1/product/microsite_data/IONNPTPDPT232022/microsite_1715832966750/pdf/View%20Syllabus.pdf?version=1854574175	**Rs. 799/-**	**VI Sem**
5	**TCS iON NPT - Java Full Stack Developer**	**10th October, 2024**	6th November, 2024 21st December, 2024 (Applications are Open)	About the test TCS iON National Proficiency Test (TCS iON NPT) is a nation-wide assessment for jobseekers and working professionals who want to test their knowledge and skills in Java Full Stack Development. All TCS iON NPT certifications are valid for 2 years. Candidates can appear for the test ONLY In-Centre at TCS iON Authorised Exam Centres. Certification tests are conducted every 2-4 weeks. However, candidates can purchase the same TCS iON NPT Variants after 90 days of their last purchase. Candidates have the opportunity to appear for multiple attempts to improve their TCS iON NPT Score on repurchases. Candidates who score 75% and above will be awarded a TCS iON NPT - Java Full Stack Developer Expert Certificate and those who score between 50% and 74% will be awarded a TCS iON NPT - Java Full Stack Developer Pro Certificate. For the rest, only the scores will be notified. Candidates with TCS iON NPT Certification can apply for jobs on the TCS iON Job Listing Portal, as well as on other job portals using this certificate.	Freshers and working professionals who would like to prove their skills in Java Full Stack Development Students who are in their pre-final or final year, pursuing any degree Anyone aspiring for job roles in Java Full Stack Development	Front-end Technology: To test the knowledge of HTML5 and CSS3, all new features in web technology, basic understanding of Javascript, jQuery Library, Javascript Frameworks like AngularJS, ReactJS, and more Design Ability: Basic prototype design and UI/UX design Version Control System(VCS): To check the knowledge of Version Control Systems like GIT, their commands and operations Server: To verify the basic knowledge of networking fundamentals, servers and server configurations Testing: To check the knowledge of Unit testing and DSA Development Language: To check the knowledge of Programming Constructs, OOP, JSP, Spring MVC Implementation, JDBC, and more **Questions - 50, Marks - 50, Time - 60 Mins**	https://www.tcsion.com/per/g01/pub/1016/iDH/instance/1/product/microsite_data/IONNPTJAVAFS09032022/microsite_1715584309852/pdf/View%20Syllabus.pdf?version=1854575437	**Rs. 799/-**	**VI Sem**
6	**TCS iON NPT - IT Web Designer**	**6th November, 2024**	To be Announced	About the test TCS iON National Proficiency Test (TCS iON NPT) is a nation-wide assessment for jobseekers and working professionals who want to test their knowledge and skills in IT Web Designing. All TCS iON NPT certifications are valid for 2 years. Candidates can appear for the test ONLY In-Centre at TCS iON Authorised Exam Centres. Certification tests are conducted every 2-4 weeks. However, candidates can purchase the same TCS iON NPT Variants after 90 days of their last purchase. Candidates have the opportunity to appear for multiple attempts to improve their TCS iON NPT Score on repurchases. Candidates who score 75% and above will be awarded a TCS iON NPT - IT Web Designer Expert Certificate and those who score between 50% and 74% will be awarded a TCS iON NPT - IT Web Designer Pro Certificate. For the rest, only the scores will be notified. Candidates with TCS iON NPT Certification can apply for jobs on the TCS iON Job Listing Portal, as well as on other job portals using this certificate.	Freshers and professionals with up to 2 years of work experience Students who are in their pre-final or final year, pursuing any degree Anyone aspiring for job roles related to IT Web Designing in top corporates	HTML Building Components - Basic HTML controls and structure New Features of HTML5 - Handling HTML5 complex components CSS Properties - Web application look and feel will be handled by CSS properties and Knowledge of CSS Properties CSS Component - CSS Properties can be applied to the HTML elements. CSS component helps programmer to perform actions on HTML components. Helps in enhancing application performance Java - Focussed on back end service implementations. Core Java, Advanced Java and Framework knowledge introduced in this section DevOps - Application Deployment Process, Configuration, Testing and Performance Management incorporated **Questions - 50, Marks - 50, Time - 60 Mins**	https://www.tcsion.com/per/g01/pub/1016/iDH/instance/1/product/microsite_data/IONNQTITWEBD01112021/microsite_1715752273128/pdf/View%20Syllabus.pdf?version=1854573666	**Rs. 799/-**	**VI Sem**
7	**TCS iON NPT - Artificial Intelligence**	**6th November, 2024**	To be Announced	About the test TCS iON National Proficiency Test (TCS iON NPT) is a nation-wide assessment for jobseekers and working professionals who want to test their knowledge and skills in Artificial Intelligence (AI). All TCS iON NPT certifications are valid for 2 years. Candidates can appear for the test ONLY In-Centre at TCS iON Authorised Exam Centres. Certification tests are conducted every 2-4 weeks. However, candidates can purchase the same TCS iON NPT Variants after 90 days of their last purchase. Candidates have the opportunity to appear for multiple attempts to improve their TCS iON NPT Score on repurchases. Candidates who score 75% and above will be awarded a TCS iON NPT - Artificial Intelligence Expert Certificate and those who score between 50% and 74% will be awarded a TCS iON NPT - Artificial Intelligence Pro Certificate. For the rest, only the scores will be notified. Candidates with TCS iON NPT Certification can apply for jobs on the TCS iON Job Listing Portal, as well as on other job portals using this certificate.	Freshers and professionals with up to 2 years of work experience Students who are in their pre-final or final year, pursuing any degree Anyone aspiring for Artificial Intelligence (AI) related job roles in top corporates Get Started	Introduction to Artificial Intelligence - Basic concepts of Artificial Intelligence (AI) Search - Focusses on search algorithms which forms the base of Artificial Intelligence (AI) Reasoning - Focusses on reasoning issues like logical reasoning which is beneficial from Artificial Intelligence point of view Machine Learning - Focusses on Machine Learning concepts and types of Machine Learning Deep Learning - Deep Learning is a section of Machine Learning. This module focusses on the concepts of Deep Learning Time Series - Time series is a pattern which is commonly followed in Artificial Intelligence (AI). This section focusses on implementation of Time Series TensorFlow - TensorFlow is a Deep Learning framework created by Google and this module focusses on the basics and terminologies of Tensorflow **Questions - 50, Marks - 50, Time - 60 Mins**	https://www.tcsion.com/per/g01/pub/1016/iDH/instance/1/product/microsite_data/IONNQTARTINT240621/microsite_1715751828256/pdf/View%20Syllabus.pdf?version=1854577370	**Rs. 799/-**	**VI Sem**

TCS Assessments (for CSE / IT / EC / EX) - VI Sem : Software Domain

TCS iON - PAID ASSESSEMENT

https://www.tcsion.com

Note: The courses/assessment available in different platforms(free/paid) are listed here as suggestion.Other courses(important from placement/skill point of view) can also be incorporated

SNo	Test Name	Next Test Date	Upcoming Test Date	About the Test	How Can Apply	Part A - Test of knowledge	Syllabus	Amount/ Fee	Year/Semester, (can be suitable for)
8	**TCS iON NPT - Machine Learning**	**6th November, 2024**	To be Announced	About the test TCS iON National Proficiency Test (TCS iON NPT) is a nation-wide assessment for jobseekers and working professionals who want to test their knowledge and skills in Machine Learning. All TCS iON NPT certifications are valid for 2 years. Candidates can appear for the test ONLY In-Centre at TCS iON Authorised Exam Centres. Certification tests are conducted every 2-4 weeks. However, candidates can purchase the same TCS iON NPT Variants after 90 days of their last purchase. Candidates have the opportunity to appear for multiple attempts to improve their TCS iON NPT Score on repurchases. Candidates who score 75% and above will be awarded a TCS iON NPT - Machine Learning Expert Certificate and those who score between 50% and 74% will be awarded a TCS iON NPT - Machine Learning Pro Certificate. For the rest, only the scores will be notified. Candidates with TCS iON NPT Certification can apply for jobs on the TCS iON Job Listing Portal, as well as on other job portals using this certificate.	Freshers and professionals with up to 2 years of work experience Students who are in their pre-final or final year, pursuing any degree Anyone aspiring for Machine Learning related job roles in top corporates	Basics of Statistics - Basic concepts of Statistics Basics of Machine Learning - Focusses on the terminologies of Machine Learning and Algorithms Machine Learning Methodology (CRISP DM) - Covers all methodologies of Machine Learning Key concepts of Machine Learning - Covers all the concepts of Machine Learning Machine Learning Algorithms with real-life use cases - Covers all the real-life use cases Exploratory Data Analysis - Exploratory data analysis (EDA) is an approach to analysing data sets to summarise their main characteristics, often with visual methods **Questions - 50 Marks - 50 Time - 60 Mins** Part B - Test of application, Data Wrangling and EDA, Python for Machine Learning, Supervised Machine Learning Algorithms with real life use cases	https://www.tcsion.com/per/pr01/pub/1016/IDH/instance/1/product/microsite_data/IONNQTMLEARN230621/microsite_1715772429944/pdf/View%20Syllabus.pdf?version=1854577445	**Rs. 799/-**	**VI Sem**